CURRICULUM VITAE

Taewoo Jo 조태우

Ph.D. Student, Department of Computer Science
Computer Graphics and Applications Lab (CGnA), Yonsei University, Seoul, Republic of Korea

twj5349@yonsei.ac.kr · twj5349@gmail.com · carvenist.github.io

RESEARCH INTERESTS

Extended Reality (XR) · Human-Computer Interaction (HCI) · Eye Tracking & Gaze Interaction · Accessibility · User Interface & User Experience

EDUCATION

2024 – Present
Ph.D. in Computer Science
Yonsei University — Computer Graphics and Applications Lab (CGnA)

2022 – 2024
M.S. in Intelligent Robotics
Gwangju Institute of Science and Technology (GIST)

2017 – 2022
B.S. in Physics
Gwangju Institute of Science and Technology (GIST)

Mar – Jun 2020
Exchange Student
Korea Advanced Institute of Science and Technology (KAIST)

RESEARCH EXPERIENCE

2024 – Present
Graduate Researcher
Computer Graphics and Applications Lab (CGnA), Yonsei University

- Bayesian modeling of implicit selection intention for manual-free gaze interaction in XR
- Non-Euclidean portal reset for continuous walking in room-scale redirected walking
- Mirror-based avatar transition visualization during VR scene changes

2020 – 2024
Research Assistant / Research Intern
Human-Centered Intelligent Systems Lab (HCIS), GIST

- WatchCap: wearable compensatory head-movement stimulation for people with low vision
- XAI visualization for autonomous vehicles — joint research with MIT CSAIL
- In-car XR platform: vehicle tracking (GPS/IMU), in-vehicle HMD tracking, driving-event-aware content
- VR-based low-vision simulation and mobile wayfinding interfaces

Jul – Aug 2021
G-SURF Research Internship
School of Integrated Technology, GIST

PUBLICATIONS

Bold denotes my name. * denotes equal contribution.

International Journal Papers

[J1] Ho Jung Lee, **Taewoo Jo**, Sulim Chun, In-Kwon Lee. Can't Nobody Stop Me! Non-Euclidean Portal Reset for Continuous Walking in Virtual Reality. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 32(5), 2026. Presented at IEEE VR 2026.

[J2] **Taewoo Jo**, Ho Jung Lee, Sulim Chun, In-Kwon Lee. Manual-Free Gaze Interaction via Bayesian-Based Implicit Intention Prediction. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 31(12), 2025. Presented at IEEE ISMAR 2026 (invited TVCG paper).

[J3] Gwangbin Kim, Dohyeon Yeo, **Taewoo Jo**, Daniela Rus, Seungjun Kim. Decoding Your Ride: What Information Builds Human Trust on Real Roads? *GetMobile: Mobile Computing and Communications*, 29(2), 2025. Invited

[J4] **Taewoo Jo**, Dohyeon Yeo, Gwangbin Kim, Seokhyun Hwang, Seungjun Kim. WatchCap: Improving Scanning Efficiency in People with Low Vision through Compensatory Head Movement Stimulation. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. (IMWUT)*, 8(2), Article 47, 2024. Presented at ACM UbiComp/ISWC 2024.

[J5] Hwaseung Jeon*, **Taewoo Jo***, Dohyeon Yeo*, Eunsol An, Yumin Kang, Seungjun Kim. The Way of Water: Exploring the Role of Interaction Elements in Usability Challenges with In-Car VR Experience. *Virtual Reality*, 28(3), 2024.

[J6] Gwangbin Kim, Dohyeon Yeo, **Taewoo Jo**, Daniela Rus, Seungjun Kim. What and When to Explain? On-Road Evaluation of Explanations in Highly Automated Vehicles. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. (IMWUT)*, 7(3), Article 104, 2023. Presented at ACM UbiComp/ISWC 2023. Distinguished Paper

[J7] **Taewoo Jo**, Dohyeon Yeo, Seungjun Kim. Enhancing Wayfinding Experience in Low-Vision Individuals through a Tailored Mobile Guidance Interface. *Electronics*, 12(22), 4561, 2023.

International Conference Papers

[C1] Dong-Yun Kim, Ho Jung Lee, **Taewoo Jo**, In-Kwon Lee. Mirror-Based Avatar Transition Visualization During VR Scene Changes. *IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, 2026. Poster

Domestic Conference Papers

[D1] **Taewoo Jo**, Dohyeon Yeo, Seungjun Kim. A Study on User Experience of Mobile Wayfinding Application for Low Vision Pedestrians. *HCI Korea*, 2022. Best Paper

Under Review

One full paper each submitted to IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR 2027) and ACM CHI Conference on Human Factors in Computing Systems (CHI 2027), September 2026.

INVITED PRESENTATIONS

2026 — **Manual-Free Gaze Interaction via Bayesian-Based Implicit Intention Prediction**
IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2026) — invited TVCG journal paper presentation

ACADEMIC SERVICE

2026 — Reviewer, ACM CHI Conference on Human Factors in Computing Systems (CHI)

2026 — Reviewer, IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR)

2026 — Reviewer, ACM Symposium on User Interface Software and Technology (UIST)

Nov 2025 — Reviewer, Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. (IMWUT)

AWARDS, HONORS AND SCHOLARSHIPS

Aug 2026 — Merit Academic Paper Award, Graduate School, Yonsei University
for *Manual-Free Gaze Interaction via Bayesian-Based Implicit Intention Prediction* (IEEE TVCG)

2024 – Present — BK21 FOUR Scholarship (Full Funding), Yonsei University

2023	Distinguished Paper Award, ACM IMWUT / UbiComp 2023
2022 – 2024	GIST Scholarship (M.S., Government Supported)
2022	Best Paper Award, HCI Korea 2022
2017 – 2022	GIST Scholarship (B.S., Government Supported)

WORK EXPERIENCE

Sep – Dec 2022	**Metaverse Fellowship** RAPA × Meta
Oct 2022	**Pitch Your Lab** GIST Metaverse Research Center, ISMAR 2022
Apr – Jun 2021	**External Developer** Unity Simulation Development, Robotry Co.
Dec 2019 – Feb 2020	**Industrial-Academic Internship** CUop, CREATY design & Breez design
2019 – Present	**Senior Mentee** Hanlim-Mirae Science Camp, The Korean Academy of Science and Technology (KAST)

TEACHING EXPERIENCE

Spring 2024	Computer Graphics — Teaching Assistant, Yonsei University
Spring 2023	Human-Computer Interaction — Teaching Assistant, GIST
2021 – 2022	Mathematics in Civilization — Teaching Assistant, GIST
Spring 2021	General Physics — Teaching Assistant, GIST
Fall 2020	Introduction to Linear Algebra — Teaching Assistant, GIST

TECHNICAL SKILLS

Languages	C#, C++, Python
Development	Unity 3D, XR / VR (Meta Quest Pro, HMD tracking), eye tracking
Analysis	SPSS, JASP
Modeling	Rhino, Autodesk 360

Last updated September 2026